FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2023

TEST DATE: 25-03-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

PART – A

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A particle having charge q and mass m is projected with velocity v in uniform magnetic field B such that the angle between the velocity and magnetic field is θ. Then the radius of curvature of the path followed by the particle is (neglect gravity)

(A) $\frac{mv\cos\theta}{Bq}$ (B) $\frac{mv\cos^2\theta}{Bq}$

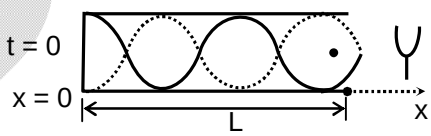
(C) $\frac{mv}{Bq\sin\theta}$ (D) $\frac{mv}{Bq\sin^2\theta}$

2. Two charge particles P and Q having masses m and 2m respectively having charge q each are released in a gravity free space when separation between then is r_0. Then speed of P after the long time is

(A) $\sqrt{\frac{kq^2}{3mr_0}}$ (B) $\frac{1}{2}\sqrt{\frac{kq^2}{3mr_0}}$

(C) $\sqrt{\frac{2kq^2}{3mr_0}}$ (D) $2\sqrt{\frac{kq^2}{3mr_0}}$

3. The equation of the standing wave (pressure wave) shown in the organ pipe is (e = end correction, f is the frequency of the tunning fork and P_0 is pressure amplitude of standing wave produced)



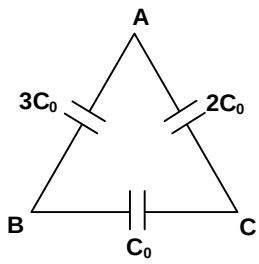
(A) $\Delta P = P_0\cos\left(\frac{7\pi x}{2(L+e)}\right)\cos(2\pi ft)$ (B) $\Delta P = P_0\cos\left(\frac{7\pi x}{2(L+e)}\right)\sin(2\pi ft)$

(C) $\Delta P = P_0\sin\left(\frac{7\pi x}{2(L+e)}\right)\cos(2\pi ft)$ (D) $\Delta P = P_0\sin\left(\frac{7\pi x}{2(L+e)}\right)\cos(2\pi ft)$

4. If $C_{eq_{AB}} = C_1$, $C_{eq_{BC}} = C_2$ and $C_{eq_{CA}} = C_3$, Then find $C_1 : C_2 : C_3$

(A) 20 : 12 : 15 (B) 3 : 5 : 4

(C) 9 : 5 : 8 (D) 1 : 2 : 3



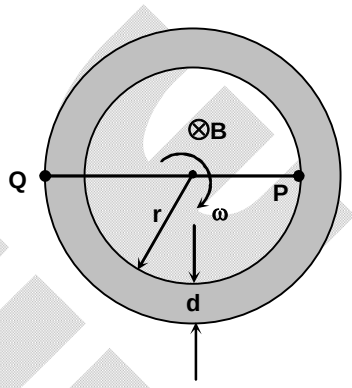
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5. A rod of mass M and length L whose linear mass density varies linearly from zero to maximum as we go away from the axis is rotating with constant angular velocity ω about the axis passing through one end and perpendicular to it. The tension in the rod at the mid-point is

(A) $\frac{M\omega^2 L}{2}$ (B) $\frac{2M\omega^2 L}{3}$

(C) $\frac{7M\omega^2 L}{12}$ (D) $\frac{7M\omega^2 L}{24}$

6. Thin circular conducting annular disc is rotating with constant angular velocity ω in the uniform magnetic field B as shown. Find potential difference between the points P and Q is (Assume d << r)



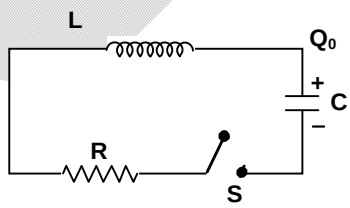
(A) $\frac{B\omega rd}{2}$

(B) $B\omega rd$

(C) $B\omega\left(\frac{r^2-d^2}{2}\right)$

(D) $B\omega(r^2-d^2)$

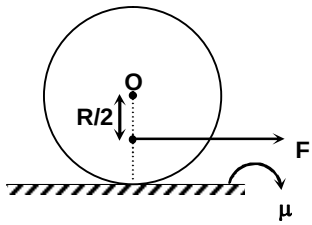
7. In the circuit shown, charge on the capacitor is Q_0 when switch is open. Now the switch is closed at t = 0, then the charge on capacitor as a function of time is



(A) $Q_0 e^{-\frac{Rt}{L}} \cos\left(\sqrt{\frac{1}{LC}+\frac{R^2}{4L^2}}\,t\right)$ (B) $Q_0 e^{-\frac{Rt}{2L}} \cos\left(\sqrt{\frac{1}{LC}+\frac{R^2}{4L^2}}\,t\right)$

(C) $Q_0 e^{-\frac{Rt}{L}} \cos\left(\sqrt{\frac{1}{LC}-\frac{R^2}{4L^2}}\,t\right)$ (D) $Q_0 e^{-\frac{Rt}{2L}} \cos\left(\sqrt{\frac{1}{LC}-\frac{R^2}{4L^2}}\,t\right)$

8. A uniform disc (m, R) is purely translating on rough horizontal surface as shown in the figure. Then the acceleration of the disc is



(A) μg (B) $\frac{\mu g}{2}$

(C) $\frac{\mu g}{3}$ (D) $\frac{2\mu g}{3}$

9. A soap bubble of radius R and thickness d(<<R) collapse and a drop is formed. Then the ratio of excess pressure inside the bubble to the drop is

(A) $2\left(\frac{3d}{R}\right)^{2/3}$ (B) $\left(\frac{d}{R}\right)^{2/3}$

(C) $2\left(\frac{3d}{R}\right)^{1/3}$ (D) $2\left(\frac{d}{R}\right)^{1/3}$

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10. A block of mass m is to be slide up along fixed inclined plane by applying a force F. If μ is the coefficient of friction and θ is the angle of inclination, then the minimum value of F would be

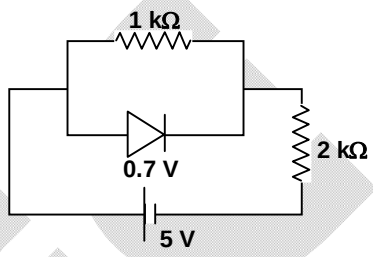
(A) $mg\frac{[\sin\theta+\mu\cos\theta]}{\sqrt{1+\mu^2}}$ (B) $mg\frac{[\sin\theta-\mu\cos\theta]}{\sqrt{1+\mu^2}}$

(C) $\mu mg\cos\theta$ (D) $mg\frac{[\mu\cos\theta-\sin\theta]}{\sqrt{1+\mu^2}}$

11. The current through the diode as shown in the figure is

(A) 0.7 mA (B) 1.45 mA

(C) 2.15 mA (D) 2.85 mA



12. The density of a uniform rod with cross section area A is d, its specific heat capacity is S and the coefficient of linear expansion is α. Calculate the amount of heat that should be added to increase the length of the rod by ℓ.

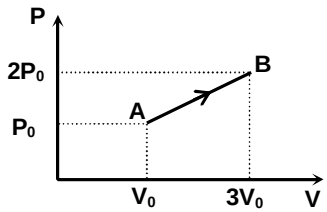
(A) $\frac{dAs\ell}{\alpha}$ (B) $\frac{\alpha As\ell}{d}v$

(C) $\frac{ds\ell}{A\alpha}$ (D) $\frac{dAs\ell}{\alpha}v$

13. One mole of a monoatomic gas undergoes a process in a given P-V diagram then average specific heat for the process is

(A) $\frac{9R}{10}$ (B) $\frac{18R}{10}$

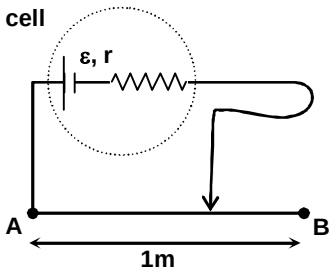
(C) $\frac{21R}{10}$ (D) $\frac{13R}{10}$



14. In the following circuit, resistance of potentiometer wire is 10Ω. Power consumption in potentiometer wire is same when jockey is 10 cm either from end A or end B. Then internal resistance of cell is

(A) 2Ω (B) 6Ω

(C) 5Ω (D) 3Ω



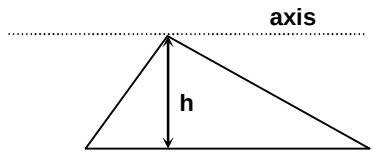
15. If minimum deviation through a prism is equal to its refracting angle of prism. Then find the refracting angle if n is the refractive index of prism.

(A) $\sin^{-1}\left(\frac{n}{2}\right)$ (B) $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$

(C) $2\sin^{-1}\left(\sqrt{\frac{1-n^2}{2}}\right)$ (D) $2\cos^{-1}\left(\frac{n}{2}\right)$

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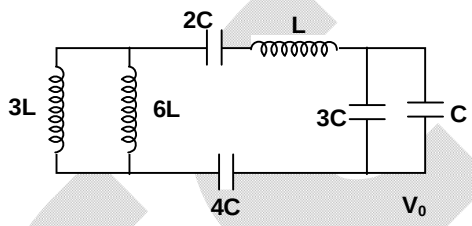
16. The moment of inertia of the uniform triangular plate about the axis (in the plane of plate) shown in the figure is (m is the total mass of plate)



(A) $\frac{mh^2}{3}$ (B) $\frac{mh^2}{2}$

(C) $\frac{2mh^2}{3}$ (D) $\frac{mh^2}{6}$

17. Frequency of the L-C oscillation shown in the diagram is



(A) $\frac{1}{2\pi\sqrt{LC}}$

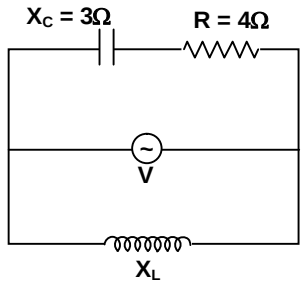
(B) $\frac{1}{2\pi\sqrt{6LC}}$

(C) $\frac{1}{2\pi\sqrt{4LC}}$

(D) $\frac{1}{2\pi\sqrt{3LC}}$

18. The 3rd excitation potential for He^+ ion in ground state is
(A) 48.36 V (B) 1.51 V
(C) 0.85 V (D) 51 V

19. If current through the source and voltage applied are in same phase, then the value of x_L is



(A) 5Ω (B) 7Ω

(C) $\frac{25}{4}\Omega$ (D) $\frac{25}{3}\Omega$

20. Two ideal gas are mixed in adiabatic chamber, then the entropy of the system will
(A) increase (B) decrease
(C) remains same (D) can't be predicted

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. Bullets of mass 10 gm each are fired from a machine gun at rate of 60 bullets/minute. The muzzle velocity of bullet is 100 m/s. The thrust force due to firing bullets experienced by the person holding the gun stationary is x newton. Find the value of x.

22. The escape velocity from the centre of earth is $\sqrt{\frac{xGM}{R}}$. Find the value of x. (The terms have their usual meanings)

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23. Work done by the force $\vec{F} = x^2y\hat{i} + \frac{x^3}{3}\hat{j}$ to move a particle from (0, 0) to (3, 9) along the curve $y = x^2$ is k joule. Find the value of k.

24. In an α-decay, the kinetic energy of α-particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the parent nucleus is (assume daughter nucleus is in ground state)

25. A particle executing SHM from x = –A to x = +A with time period T. The shortest time taken by the particle to reach from $x = \frac{A}{2}$ to $x = \frac{-\sqrt{3}A}{2}$ is $\frac{T}{m}$. Find the value of m.

26. The focal length of a thin biconvex lens is 20 cm. When an object is moved from a distance of 25 cm to 50 cm in front of it, the magnification changes from m_1 to m_2. Find $\frac{m_1}{m_2}$.

27. A ball is dropped from a height. If it takes 0.200 sec to cross the last 6.00 m before hitting the ground. Find the height (in m) from which it was dropped. ($g = 10\ m/s^2$)

28. 7th order minima is observed at central point O in case of YDSE if a glass slab of thickness t is placed infront of one of the slit. The thickness of glass slab is $n \times 10^{-7}$ m. Find n. (Given $\mu_g = \frac{3}{2}$ and $\lambda = 6000\ \text{Å}$)

29. If the measured values of a, b, c, d are $20 \pm 0.2, 10 \pm 0.4, 10 \pm 0.1$ and 10 ± 0.2. Then the maximum percentage error in $\frac{(a+b)c}{d}$ is

30. Velocity of a particle moving along x-axis is given as $v = 2t - t^2$, then the distance traveled by particle from t = 0 to t = 4 sec is x meter. Find the value of x.

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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. Which of the following molecules will give Seliwanoff's test?
(A) Fructose (B) Glycine
(C) Starch (D) Galactose

32. Which are the monomers of Kevlar?
(A) Phenol and formaldehyde
(B) Aniline and terephthaloyl chloride
(C) 1, 4-phenylene diamine and benzoyl chloride
(D) 1,4-phenylene diamine and terephthaloyl chloride

33. ZnO shows yellow colour on heating due to
(A) d – d transition
(B) charge transfer spectra
(C) higher polarization caused by Zn^{2+}
(D) presence of F-centre

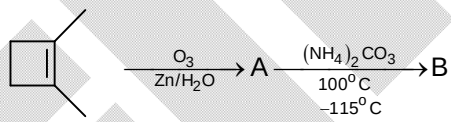
34. Arrange according to increasing order of s-character of central element.

XeF_2, XeO_3, $XeOF_4$, $\overset{\oplus}{N}O_2$

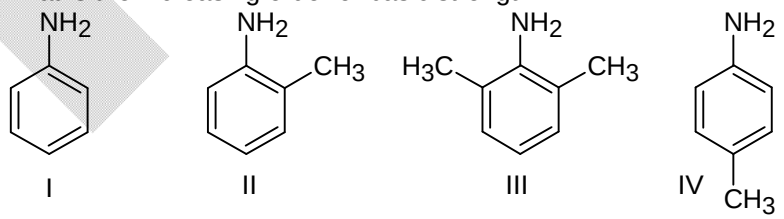
I II III IV

(A) II < IV < III < I (B) IV < III < I < II
(C) II < I < IV < III (D) III < I < II < IV

35. Which of the following statement is incorrect?
(A) Li and Mg both are capable of formation of organometallic compounds.
(B) Li and Mg both form stable nitrides.
(C) Be and Al both react with concentrated NaOH to form H_2.
(D) Be and Al both form polymeric chloride in the solid state.

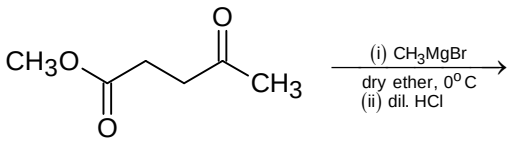
36. 

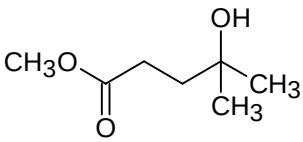
(A) It is five membered ring. (B) It is non-aromatic.
(C) It is aromatic. (D) It is anti-aromatic.

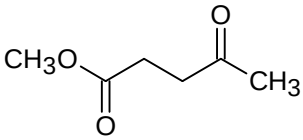
37. What is the increasing order of basic strength?


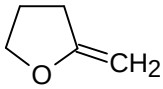
I II III
(A) I < II < IV < III (B) II < IV < I < III
(C) I < II < IV < III (D) III < II < I < IV

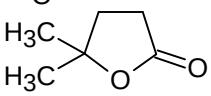
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38. 

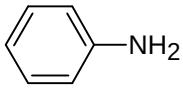
(A) 

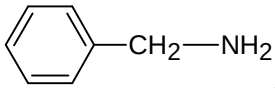
(B) 

(C) 

(D) 

39. Which of the amines can not be prepared by Gabriel phthalimide synthesis?

(A) 

(B) 

(C) $CH_3 - CH_2 - NH_2$

(D) $CH_3 - NH_2$

40. Calculate the approximate pH of the resulting solution formed by mixing up of 25 ml 0.04 M Na_2CO_3 and 20 ml 0.05 M HCl. $[pK_1 = 6.4$ and $pK_2 = 10.3$ for $H_2CO_3]$

(A) 10.5 (B) 8.35
(C) 6.88 (D) 5.15

41. If is given that

	E^o
$Ca^{2+} + 2e \longrightarrow Ca$	−2.87 volt.
$Na^{+} + e \longrightarrow Na$	−2.71 volt.
$Fe^{2+} + 2e \longrightarrow Fe$	−0.44 volt.
$Li^{+} + e \longrightarrow Li$	−3.04 volt.

What is the order of reducing character?

(A) Li < Na < Fe < Ca (B) Fe < Ca < Na < Li
(C) Fe < Na < Li < Ca (D) Fe < Na < Ca < Li

42. CFSE of the complex $[FeL_6]^{2-}$ is $-2.4\Delta_o + 3P$

What is the d-electron configuration?

(A) $t_{2g}^6 e_g^1$ (B) $t_{2g}^6 e_g^0$
(C) $t_{2g}^4 e_g^1$ (D) t_{2g}^5

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43. $K_2Cr_2O_7$ on heating with KCl and concentrated H_2SO_4 gives a brown fume (A), which on reaction with NaOH gives a yellow solution. What is the number of O atoms present in one molecule of 'A'?

(A) 2 (B) 3
(C) 1 (D) 4

44. Match List-I with List-II

List – I (Reactions)		**List – II Enzyme**
(P)	$C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$ (Glucose)	(I) Invertase
(Q)	$2(C_6H_{10}O_5)_n(aq) + nH_2O(\ell) \longrightarrow nC_{12}H_{22}O_{11}(aq)$ Starch Maltose	(II) Zymase
(R)	$C_{12}H_{22}O_{11} + H_2O(\ell) \longrightarrow 2C_6H_{12}O_6$ Maltose Glucose	(III) Diastase
(S)	$C_{12}H_{22}O_{11}(aq) + H_2O(\ell) \longrightarrow C_6H_{12}O_6(aq) + C_6H_{12}O_6(aq)$ Sucrose Glucose Fructose	(IV) Maltase

(A) P – III; Q – II; R – I, S – IV (B) P – I; Q – II; R – III, S – IV
(C) P – IV; Q – II; R – I, S – III (D) P – II; Q – III; R – IV, S – I

45. If the pressure and density of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$ in adiabatic process changes from (p, d) to (p′, d′), when p = 32 p′, then $\frac{d}{d'}$ is equal to:

(A) $\frac{1}{128}$ (B) $\frac{25}{27}$
(C) 128 (D) $\frac{7}{39}$

46. PCl_3 on hydrolysis gives an acid 'A', which can reduce $AgNO_3$ to metallic silver. Identify 'A'.

(A) H_3PO_4 (B) H_3PO_2
(C) $H_2P_4O_7$ (D) H_3PO_3

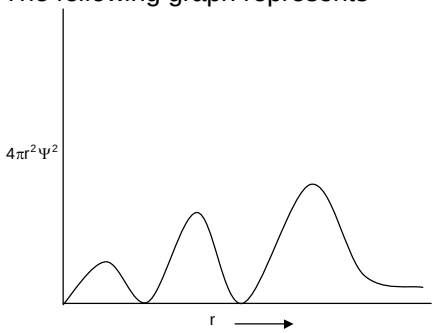
47. Match List I with List II

List I (Metals)		**List II (Method of refining)**	
(A)	Copper	I.	van Arkel method
(B)	Zinc	II.	Electrolytic refining
(C)	Zirconium	III.	Distillation
(D)	Indium	IV.	Zone refining

(A) A – II; B – III; C – I; D – IV
(B) A – III; B – IV; C – II; D – I
(C) A – I; B – II; C – IV; D – III
(D) A – II; B – I; C – III; D – IV

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48. The following graph represents



(A) 2P orbital
(B) 3S orbital
(C) 2S orbital
(D) 3P orbital

49. The activation energies of the first order reactions R_1, R_2, R_3 and R_4 are 40 kJ/mol. 35 kJ/mol, 55 kJ/mol and 25 kJ/mol respectively. What is the increasing order of their temperature dependency?

(A) $R_1 < R_2 < R_3 < R_4$
(B) $R_2 < R_1 < R_4 < R_3$
(C) $R_4 < R_2 < R_1 < R_3$
(D) $R_2 < R_3 < R_4 < R_1$

50. The density of a liquid is 1.2 gm/ml. There are 35 drops in 2 ml. Find out the number of molecules in one drop: (Molecular weight of liquid = 70) [N_A = Avogadro's number]

(A) $\frac{1.2}{35}N_A$
(B) $\left(\frac{1}{35}\right)^2 N_A$
(C) $\frac{1.2}{(35)^2}N_A$
(D) $1.2N_A$

SECTION – B

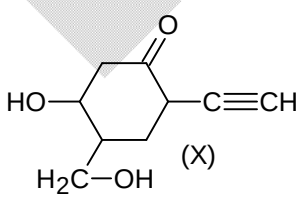
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. If the temperature of an ideal gas is increased by four times and the pressure is increased by two times, the mean free path of the gas is increased by 'x' times. Find x = ?

52. $XeF_4 + 36F_5 \longrightarrow A(\text{cation}) + B(\text{anion})$
Find out number of lone pairs in A.

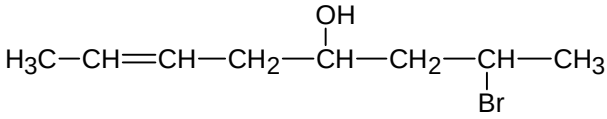
53. How many effective Na^+ ions are present in the rock salt NaCl, if the cations along one axis joining opposite face centres are removed?

54. 

(X)
If 'n' is the number of moles of CH_3MgBr are consumed by one mole of the given molecule X, then 'n' = ?

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55. What is the total number of stereoisomers formed by following molecule?



56. K_f of water is 1.86 K kg mol^{-1}. If a radiator can hold maximum 1 kg of water, how many grams of ethylene glycol $(C_2H_6O_2)$ must be added to lower the freezing point of solution to -2.8^oC.

57. If the total number of electrons lost by Cu_2S is x, then x = ?

$$Cu_2S + 2O_2 \longrightarrow 2CuO + SO_2$$

58. $2NO_2(g) + O_3(g) \longrightarrow N_2O_5(g) + O_2(g)$

The reaction takes place through following mechanism

$O_3(g) \rightleftharpoons O_2(g) + O(g)$ Fast

$NO_2(g) + O(g) \longrightarrow NO_3(g)$ Slow

$NO_2(g) + NO_3(g) \longrightarrow N_2O_5(g)$

What is the overall order of reaction?

59. How many moles of phosphine gas are formed when 3 moles of calcium phosphide react with excess water?

60. Find out number of unpaired electrons in t_{2g} orbital of cobalt in $[CoF_6]^{3-}$

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. The complete set of real values of 'a' $\in R^+$ for which both roots of the equation $x^2 + 2ax + 9 = 0$ lies in (–6, 1) is

(A) $\left[3, \frac{15}{4}\right)$ (B) $\left(\frac{39}{4}, 12\right)$

(C) [9, 12) (D) $(-2, 0) \cup (2, 9)$

62. For any natural number n > 1, consider the sum $S = \frac{1}{n^2+1} + \frac{2}{n^2+2} + ... \frac{2n}{n^2+2n}$, then S_∞ is equal to

(A) $\frac{1}{2}$ (B) 2

(C) 4 (D) $\frac{1}{4}$

63. The remainder obtained when $10^{2023} + 12^{2023}$ is divided by 121 is equal to

(A) 2 (B) 8

(C) 9 (D) 3

64. Using the letters of word **SANATAN**, words are from such that all A are together is m and no two A are together is n, then number of divisor of M × N is

(A) 10 (B) 24

(C) 54 (D) 36

65. Two cards are drawn from each of the 3 well shuffled pack of 52 cards. Then the probability that exactly one card is identical among the cards drawn is equal to

(A) $\frac{4 \cdot {}^{50}C_2}{\left({}^{52}C_2\right)^2}$ (B) $\frac{2 \cdot {}^{50}C_2}{\left({}^{52}C_2\right)^2}$

(C) $\frac{6 \cdot {}^{50}C_2}{\left({}^{52}C_2\right)^2}$ (D) $\frac{12 \cdot {}^{50}C_2}{\left({}^{52}C_2\right)^3}$

66. If matrix $M_n = \begin{bmatrix} n+1 & n-2 \\ n+2 & n \end{bmatrix}$, $n \in N$, then $\frac{\sum_{r=1}^{2023} |M_n|}{2023}$ is equal to (where $|M_n|$ = det. M_n)

(A) 2023 (B) 1012

(C) 1016 (D) 2024

67. The value of $\sum_{r=0}^{20} (-1)^r \left({}^{20}C_r\right)^2 r(20-r)$ is equal to

(A) $400.{}^{38}C_9$ (B) $-40.{}^{38}C_9$

(C) $400.{}^{40}C_9$ (D) none of these

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68. If the tangents AP and AQ are drawn from the point A(8, –1) to the circle $x^2 + y^2 - 3x + 2y - 7 = 0$, C be the centre then area of quadrilateral APCQ is

(A) $2\sqrt{41}$ (B) $3\sqrt{41}$

(C) $2\sqrt{82}$ (D) $3\sqrt{82}$

69. The function F : [0, 3] $\rightarrow$ [5, 33] defined by $f(x) = 2x^3 - 15x^2 + 36x + 5$, is

(A) one-one and onto (B) onto but not one-one

(C) neither one-one nor onto (D) one-one but not onto

70. Let m and n be two positive integer greater than 1. If $\lim\limits_{\alpha\to 0}\left(\frac{e^{\cos\alpha^h} - e}{\alpha^k}\right) = -\frac{e}{2}$, then locus of P(h, k) is

(A) straight line with slope 2 (B) straight line with slope 3

(C) circle of radius 2 (D) circle of radius 3

71. Let y(x) be a solution of the differential equation $(1 + e^x)\, y' + ye^x = 1$, if y(0) = 0, then which of the following is correct ?

(A) y(–4) = 0 (B) y(–2) = 0

(C) y(x) has a critical point in (–1, 0) (D) y(x) has a critical point in (0, 2)

72. Let 3, 7 – a and $\frac{10 - 3b^2}{b + 2}$ be three natural number, which are 1st three terms of an A.P. respectively. If (b, a) $\in$ I, then the number of possible such A.P is

(A) 2 (B) 3

(C) 4 (D) 5

73. Let f : (0, 1) $\rightarrow$ R be a function defined by $f(x) = 10x^{\sin x + \cos x}$. If $I = \int_0^1 f(x)\,dx$, then number of value of $[I^2]$ is equal to

(A) 8 (B) 9

(C) 10 (D) 11

74. An ellipse is orthogonal to the hyperbola $x^2 - y^2 = 2$, the eccentricity of the ellipse is reciprocal of the hyperbola, then

(A) equation of ellipse is $2x^2 + y^2 = 8$

(B) focus of ellipse is at $\left(0, 4\sqrt{2}\right)$

(C) equation of directrix is $y + 4\sqrt{2} = 0$

(D) equation of director circle of ellipse is $x^2 + y^2 = 12$

75. If f(0) = 3 and $\lim\limits_{t\to x}\frac{\sec x\, f(t) - f(x)\sec t}{t - x} = \sec^2 x$, then the value of f'(0) + f'(π) is

(A) 1 (B) 0

(C) 2 (D) 3

76. The normal to the curve (x – 2) (y + 2) = 4 at the point (3, 2) meets the curve again at

(A) (–2, –3) (B) $\left(-6, \frac{3}{2}\right)$

(C) $\left(-14, -\frac{9}{4}\right)$ (D) (1, –2)

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77. The director circle of the parabola $(y-2)^2 = 16(x+7)$ touches the circle $(x-1)^2 + (y+1)^2 = r^2$, at
(A) (1, 11) (B) (–1, –11)
(C) (–11, –1) (D) (–11, 1)

78. If $3^x = 4^y = 5^z = 60^{-\omega}$, then the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}+\frac{1}{\omega}$ is equal to
(A) 11 (B) 3
(C) $\frac{7}{2}$ (D) 0

79. The lines with vector equation are $\vec{r}_1 = -3\hat{i}+6\hat{j}+\lambda\left(-4\hat{i}+3\hat{j}+2\hat{k}\right)$ and $\vec{r}_2 = -2\hat{i}+7\hat{j}+\mu\left(-4\hat{i}+\hat{j}+\hat{k}\right)$ are such that
(A) they are coplanar (B) they are intersection
(C) distance between line is $\frac{1}{3}$ units (D) angle between them is $\tan^{-1}\left(\frac{4}{3}\right)$

80. If the function $2x^3 - (8-a)x^2 + \left(a^2+\frac{16}{9}\right)x - 12$ has a local minima at some $x \in R^-$, then number of integral value of a is
(A) 2 (B) 4
(C) 7 (D) 0

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 6 \\ 3 & 4 & 8 \end{bmatrix}$, then $|adj(adj(2A))| = 2^{\lambda}$, then λ is equal to

82. If Q be the image of point P(1, –2, 3) with respect to plane $x - y + z = 7$. The distance of Q from origin is d, then [d] is

83. If f(x) is continuous and derivable on [–3, 6] and $-4 \le f'(x) \le 3\ \forall\ x \in (-3, 6)$. Then difference of maximum and minimum value of f(6) is equal to

84. If $S_n = \sum_{r=1}^{n} \cot^{-1}\left(r^2+3r+3\right)$, then $2023\tan(S_6)$ is equal to

85. If f(x) is a monic polynomial function of degree 4 satisfying $f(i) = \frac{1}{i}$ for i = 1, 2, 3, 4, then number of zeros at the end of f(5) is

86. Let $f(x) = \int_0^{\pi} \ln\left(1-2x\cos\theta+x^2\right)d\theta$, then $\frac{f(49)}{f(7)}$ is equal to

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87. Let $z \in c$ satisfying the condition $|z + 1|^2 + |z - 1|^2 + 2|z| = 6$, then maximum value of $|z - 3 - 4i|$ is equal to

88. Let number of real and non real roots of equation $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ are m and n respectively, then the value of $(m + 1)^n$ is equal to

89. Let $\vec{a},\vec{b},\vec{c}$ be three vectors of magnitude 2, 3 and 4 respectively, satisfying $\left|\left[\vec{a}\,\vec{b}\,\vec{c}\right]\right| = 24$. Then the value of $\left(2\vec{a}+\vec{b}+\vec{c}\right)\cdot\left(\left(\vec{a}\times\vec{b}\right)\times\vec{b}+6\vec{c}\right)$ is equal to

90. If A, B, C are the angles of triangle ABC and tanA, tanB, tanC are the roots of the equation $x^4 - 3x^3 + 3x^2 + 2x + 5 = 0$, then if its fourth root is k, then the value of 3k is

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